GOVERNMENT OF KARNATAKA

DEPARTMENT OF SCHOOL EDUCATION (PRE UNIVERSITY)

Lesson Based Assessment UNIT TEST – 1

2nd PU PHYSICS (33)

Unit 1 Chapter: Electric Charges and Fields

Maximum Marks: 30 **Total Time**: 1 Hour

General Instructions:

1. All parts (A to D) are compulsory.
2. For Part – A questions, only first written-answer will be considered for evaluation.
3. Answers without relevant diagram/figure/circuit wherever necessary will not carry any marks.
4. Direct answers to numerical problems without relevant formula and detailed solutions will not carry any marks.

Part A

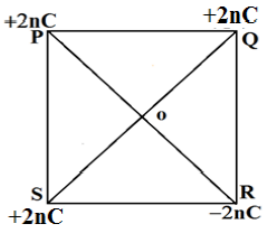
I] Choose the correct answer from the given options **7 × 1 = 7**

1. Dipole moment is a vector quantity which is pointing from
a. -q to +q along the axis of an electric dipole
b. +q to -q along the axis of an electric dipole
c. Infinity to +q
d. +q to infinity

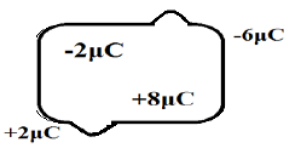
2. Electric field intensity at a distance 'r' from the centre of a uniformly charged sphere outside the sphere is 'E'. Which of the following is correct?
a. $E \propto q$ and $E \propto 1/r$
b. $E \propto 1/q$ and $E \propto 1/r$
c. $E \propto q$ and $E \propto 1/r^2$
d. $E \propto q$ and $E \propto 1/r^3$

3. The physical quantity with SI unit of C/m is:
a. Volume charge density
b. Surface charge density
c. Linear charge density
d. Electric dipole moment

4. Four charges are placed at the corners of a square PQRS as shown in the figure. The direction of the resultant electric field at the intersection of the diagonals is:
a. Towards P
b. Towards Q
c. Towards R
d. Towards S



5. The given figure is related to Gauss' law on electrostatics $\phi = q/\epsilon_0$ where q is equal to:
a. +2μC
b. -8μC
c. -6μC
d. +6μC



6. Some physical quantities are listed in Column 1 and their dimensions are listed in Column 2. Identify the correct match
a. (i) – (B), (ii) – (C), (iii) – (A)
b. (i) – (C), (ii) – (B), (iii) – (A)
c. (i) – (B), (ii) – (A), (iii) – (C)
d. (i) – (A), (ii) – (C), (iii) – (B)

Column 1	Column 2
(i) Electric Field	(A) [AT]
(ii) Electric Flux	(B) $[MLT^{-3}A^{-1}]$
(iii) Electric Charge	(C) $[ML^3T^{-3}A^{-1}]$

7. Which of the following quantity is not a basic property of electric charge?
a. Charge is quantized
b. Charge is conserved
c. Charge is a vector
d. Charge is additive

II] Fill in the blanks by choosing the appropriate answer given in the bracket for ALL of the following questions. **3 × 1 = 3**

(***energy, vacuum, positive, charge***)

8. A body gains _____ charge when it looses electron.

9. The electrostatic force between two charges is maximum in _____.

10. Gold leaf electroscope is used to detect _____ present on the body.

Part B

III] Answer any TWO of the following questions. Each question carries 2 marks. **2 × 2 = 4**

11. Write Coulomb's law in vector form and explain the terms.

12. What are electric field lines? Do electric field lines form closed loops?

13. When does an electric dipole placed in a uniform electric field experience (a) maximum torque (b) minimum torque?

Part C

IV] Answer any TWO of the following questions. Each question carries 3 marks. **2 × 3 = 6**

14. Write the three basic properties of electric charges.

15. Derive an expression for the torque experienced by an electric dipole placed in a uniform electric field.

16. Two positive charges of 0.2 μC and 0.8 μC are placed at a distance 0.15m apart. Find the position of the point on the line joining them where the electric field is zero.

Part D

V] Answer any ONE of the following questions. Each question carries 5 marks. **1 × 5 = 5**

17. Derive an expression for electric field at a point on the axial line of an electric dipole.

18. State Gauss' law in electrostatics. Using Gauss' law, derive the expression for the electric field due to an infinitely long straight uniformly charged wire.

VI] Answer any ONE of the following questions. Each question carries 5 marks. **1 × 5 = 5**

19. A pendulum bob of mass 80 mg and carrying a charge 2×10^{-8}C is at rest at a certain angle with the vertical in a horizontal uniform electric field of 20,000 V/m. Find the tension on the thread of the pendulum and the angle it makes with the vertical.

20. Two point charges each of +3μC are placed at the two corners A and B of an equilateral triangle ABC of side 0.1m. Find the magnitude and direction of the resultant electric field at C.

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